

Python Assessment

Q1. Which of the following is a valid Python variable name?

a) 1value b) value_1 c) value-1 d) @value

Q2. What will be the output of the following code? `x = 5, y = '5', print(x + y)`

a) 10 b) 55 c) TypeError d) None

Q3. Which data type is immutable in Python?

a) List b) Dictionary c) Set d) Tuple

Q4. What is the data type of `x = True`?

a) int b) bool c) str d) float

Q5. Which function is used to check the data type of a variable?

a) data() b) typeof() c) type() d) check()

Q6. What will be the output? `a = 10, b = 3, print(type(a / b))`

a) int b) float c) str d) bool

Q7. Write a Python program to swap two variables without using a third variable.

Q8. Identify the error in the code: `1name = 'Python'` and rewrite it correctly.

Q9. Write a program that assigns multiple values to multiple variables in a single line and prints them.

Q10. Write a program to convert a user-entered integer into float and string and print all values.

Q11. Explain with an example the difference between mutable and immutable data types.

Q12. Which data structure allows duplicate values?

a) Set b) Dictionary c) List d) None

Q13. What will be the output? `my_list=[1,2,3]; my_list.append([4,5]); print(len(my_list))`

a) 3 b) 4 c) 5 d) 2

Q14. Which method removes a key-value pair from a dictionary?

a) remove() b) pop() c) delete() d) discard()

Q15. What happens when you try to modify a tuple element?

a) Allowed b) No change c) TypeError d) IndexError

Q16. Which data structure does not support indexing?

a) List b) Tuple c) Set d) String

Q17. What will be the output? `my_set={1,2,3,3,4}`

a) {1,2,3,4} b) {1,2,3,3,4} c) Error d) None

Q18. Which method adds multiple elements to a set?

a) `add()` b) `append()` c) `extend()` d) `update()`

Q19. How do you access value of key 'name' from dictionary d?

a) `d.name` b) `d('name')` c) `d['name']` d) `d->name`

Q20. What will be the output? `list1=[1,2]; list2=list1; list2.append(3); print(list1)`

a) [1,2] b) [1,2,3] c) Error d) None

Q21. Which data structure is best for storing unique elements?

a) List b) Tuple c) Set d) Dictionary

Q22. What is the output? `print(5 > 3 and 2 < 1)`

a) True b) False c) Error d) None

Q23. Which operator checks both value and type?

a) `==` b) `=` c) `is` d) `!=`

Q24. What will be the output? `if 0: print('Yes') else: print('No')`

a) Yes b) No c) Error d) None

Q25. Which loop is guaranteed to execute at least once?

a) for b) while c) do-while d) None

Q26. What is the output? `for i in range(1,5,2): print(i,end=' ')`

a) 1 3 b) 1 2 3 4 c) 2 4 d) Error

Q27. What does `break` do in a loop?

a) Skip iteration b) Stop loop c) Restart loop d) End program

Q28. Which keyword skips current iteration?

a) `stop` b) `pass` c) `continue` d) `break`

Q29. What will be the output? `def fun(x): return x*x; print(fun(3))`

a) 6 b) 9 c) 3 d) Error

Q30. What happens if a function does not return anything?

a) Error b) Returns 0 c) Returns None d) Returns False

Q31. Which operator has the highest precedence?

a) `+` b) `*` c) `**` d) `//`

Q42. Write a program to check whether a number entered by the user is even or odd.

Q43. Write a program to check voting eligibility based on user age.

Q44. Write a program to take 5 numbers from the user and print maximum and minimum.

Q45. Write a program to take a string from the user and count the number of vowels.

Q46. Write a program to take a number from the user and display its multiplication table.

Q47. Write a program to accept a list of numbers from the user and print only even numbers.

Q48. Write a program to take a sentence from the user and count the number of words.

Q49. Write a function that takes two numbers from the user and returns the larger number.

Q50. Write a function that takes a list of numbers from the user and returns the sum of all elements.